

## Linux File Maintenance Commands (Core 5)

### 1 **cp** — Copy files & directories

#### ◆ Purpose

Creates a **duplicate** of a file or directory.

```
cp source destination
```

#### ◆ Common usage

```
cp file.txt backup.txt
```

```
cp file.txt /tmp/
```

```
cp -r folder1 folder2
```

#### ◆ Important options

- -r → recursive (mandatory for directories)
- -i → prompt before overwrite
- -v → verbose (show what's happening)

```
cp -riv src/ backup/
```

 Mental model:

**Original stays, copy is created**

### 2 **mv** — Move & rename

#### ◆ Purpose

Moves a file/directory **or renames it**.

```
mv source destination
```

#### ◆ Common usage

```
mv old.txt new.txt # rename
```

```
mv file.txt /home/user/ # move
```

```
mv folder1 folder2
```

#### ◆ Important options

- -i → confirm overwrite
- -v → verbose

 Mental model:

**Same file, new location or new name**

### 3 **rm** — Remove files & directories

#### ◆ Purpose

Deletes files and directories **permanently**.

```
rm file.txt
```

#### ◆ Common usage

```
rm file.txt
```

```
rm -r folder
```

```
rm -rf folder
```

#### ◆ Important options

- -r → recursive (directories)
- -f → force (no prompt)
- -i → interactive delete

## ⚠️ Danger zone

`rm -rf /`

🧠 Mental model:

**No recycle bin. No undo.**

## 4 touch — Create file / update timestamp

### ◆ Purpose

Creates an **empty file** or updates timestamps.

`touch file.txt`

### ◆ Common usage

`touch a.txt b.txt c.txt`

`touch -t 202601151030 file.txt`

🧠 Mental model:

**"Make sure this file exists."**

## 5 mkdir — Create directories

### ◆ Purpose

Creates directories.

`mkdir folder`

### ◆ Common usage

`mkdir logs`

`mkdir parent/child`

`mkdir -p parent/child/grandchild`

### ◆ Important option

- `-p` → create parent directories automatically

🧠 Mental model:

**Prepare space for files**

## 🔄 Quick Comparison (Sticky Memory)

| Command            | Action           | Original kept? |
|--------------------|------------------|----------------|
| <code>cp</code>    | Copy             | ✅ Yes          |
| <code>mv</code>    | Move / Rename    | ❌ No           |
| <code>rm</code>    | Delete           | ❌ No           |
| <code>touch</code> | Create file      | ✅ Yes          |
| <code>mkdir</code> | Create directory | ✅ Yes          |

## 🧠 Real-life flow example

`mkdir backup &&`

`cp data.txt backup/ &&`

`rm data.txt`

Meaning:

Create folder → copy file → delete original